



ST BENEDICT'S

SUBJECT	Physical Science: Paper 1	DATE	15 November 2021
GRADE	11	MARKS	200 Marks
EXAMINER	Mr D. du Toit	MODERATOR	Mrs S. Govender Mr F. McCoy
NAME	MARKING GUIDELINES	DURATION	3 hours
CLASS			

QUESTION 1

20 MARKS

1.1	C ✓✓	(2)
1.2	A ✓✓	(2)
1.3	B ✓✓	(2)
1.4	D ✓✓	(2)
1.5	B ✓✓	(2)
1.6	C ✓✓	(2)
1.7	B ✓✓	(2)
1.8	A ✓✓	(2)
1.9	B ✓✓	(2)
1.10	D ✓✓	(2)

QUESTION 2 MR NTOMBELA**25 MARKS**

2.1.1 Distance: Length of path travelled ✓

A to B 20 + B to C 36 = 56 m ✓

Displacement: change in position ✓

Straight line OR shortest distance from A to C ✓

(4)

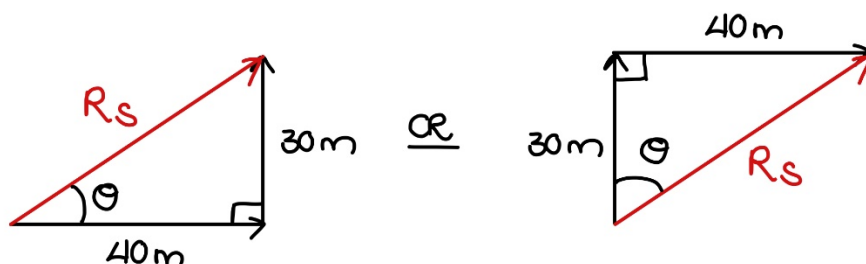
2.1.2 Rate of change of distance ✓✓AON

(2)

2.1.3 $speed = \frac{distance}{time} = \frac{56}{20} = 2,8 \text{ m} \cdot \text{s}^{-1}$ ✓

(3)

2.1.4



1 mark for resultant with arrow from tail of first vector to head of last ✓

1 mark for 90° and θ ✓

1 mark for 20 m with arrow ✓

1 mark for 30 m with arrow ✓

(4)

2.1.5 $R_s = \sqrt{40^2 + 30^2}$ ✓pythag
 $= 50 \text{ m}$ ✓

$\tan^{-1}\left(\frac{30}{40}\right) = 36,87^\circ$ ✓ or $\tan^{-1}\left(\frac{40}{30}\right) = 53,13^\circ$

$v = \frac{s}{t} = \frac{50}{20} = 2,5 \text{ m} \cdot \text{s}^{-1}$ ✓ $36,87^\circ \text{ N of E}$ ✓ OR $53,13^\circ \text{ E of N}$

(or on a bearing of $53,13^\circ$)

Mark question based on 2.1.4 diagram

(5)

2.2.1

Paul:

$$v = \frac{s}{t}$$

$$6\checkmark = \frac{s}{x + 3\checkmark}$$

$$s = 6x + 18$$

John:

$$s = ut + \frac{1}{2}at^2$$

$$s = \frac{1}{2}(1,2\checkmark)(x^2)$$

$$s = 0,6x^2$$

$$6x + 18 = 0,6x^2\checkmark$$

$$x = 12,42\text{ s}\checkmark$$

(5)

2.2.2

$$s = 6(12,42) + 18\checkmark \text{ coe 2.2.1}$$

OR

$$s = (0,6)(12,42^2)\checkmark \text{ coe 2.2.1}$$

$$s = 92,50\text{ m}\checkmark \text{ coe 2.2.1}$$

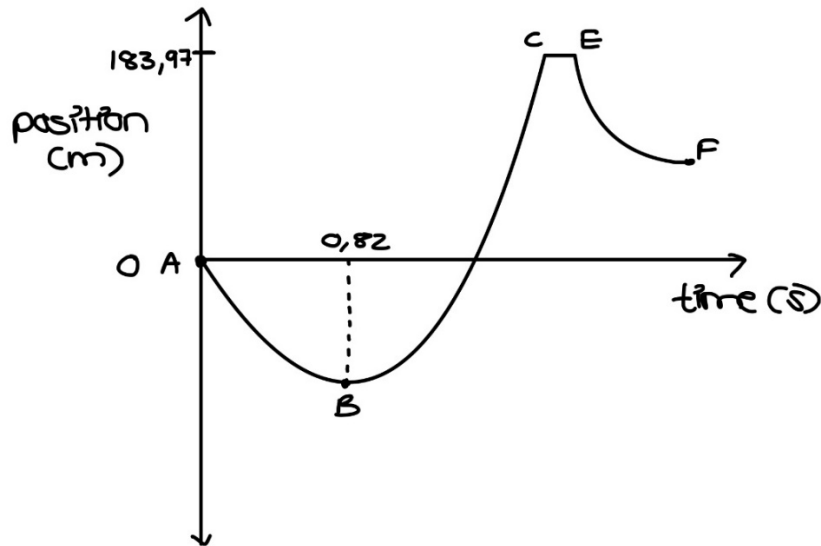
$$s = 92,50\text{ m}\checkmark \text{ coe 2.2.1}$$

(2)

QUESTION 3 MR NTOMBELA**28 MARKS**

- 3.1 Rate of change of position ✓✓ (2)
- 3.2 8 m.s^{-1} ✓ up✓ (2)
- 3.3 Decreasing velocity✓ up✓ OR uniform acceleration✓ down✓ (2)
- 3.4 $v = u + gt$
 $0 = -8✓ + 9,8✓t$
 $t = 0,82 \text{ s}✓$ (3)
- 3.5 $\text{area} = \frac{1}{2}bh = \frac{1}{2}(7 - 0,82✓\text{coe 3.4})(60,6✓) = 187,37 \text{ m}✓$
OR
 $s = ut + \frac{1}{2}gt^2$
 $s = (0)(7 - 0,82) + \frac{1}{2}(9,8)(7 - 0,82✓)^2✓$
 $= 187,37 \text{ m}✓$ (3)
- 3.6 $187,37✓\text{coe 3.5} - \frac{1}{2}(8)(0,82✓\text{coe 3.4})$
 $= 184,10 \text{ m}✓$
OR
 $s = ut + \frac{1}{2}gt^2$
 $s = (-8)(0,82) + \frac{1}{2}(9,8)(0,82)^2$
 $s = 3,27 \text{ m}✓$
 $s = 187,37✓ - 3,27 = 184,10 \text{ m}✓$ (3)
- 3.7 Maximum height ✓ after bounce ✓ (2)

3.8



✓ Values (time taken to reach max height and height of cliff)

✓ A to B shape

✓ B to C shape

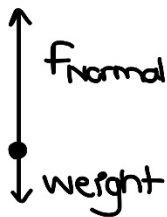
✓ E to F shape

-1 if inverted

(4)

3.9 Draw a free-body diagram showing all the forces acting on the ball at point D.

(2)



✓ weight

✓ Fnormal or Fground on balls greater than weight

3.10

$$a = \text{slope} = \frac{\Delta y}{\Delta x} \checkmark = \frac{60,6 - (-120)}{7 - 7,2} = -363 \text{ m} \cdot \text{s}^{-2}$$

$$F_{\text{net}} = ma \checkmark$$

$$= (0,6)(-363)$$

$$= -217,8 \text{ N}$$

$$F_{\text{net}} = F_N + F_g$$

$$-217,8 = F_N + (0,6)(9,8) \checkmark \checkmark$$

$$F_N = -223,68 \text{ N} \checkmark$$

QUESTION 4 MR DU TOIT**17 MARKS**

4.1



4.2 The external force acts on the truck ✓

The inertia ✓ of the block resists a change in motion ✓

The friction between the block and truck is not enough to accelerate it with the truck ✓ (4)

4.3

4.3.1 Rate of change of velocity ✓✓AON (2)

4.3.2 $v^2 = u^2 + 2as$ ✓

$0^2 = 22^2 + 2a(38,2)$ ✓

$a = 6,34 \text{ m} \cdot \text{s}^{-2}$ ✓ backwards ✓ (4)

4.3.3 $F_{net} = ma$

$F_{net} = (10\,800)(-6,34)$ ✓ = $-69769,07 \text{ N}$ ✓ (2)

4.3.4 Using rubber will increase the friction ✓

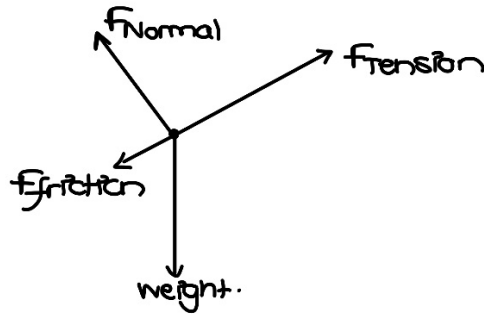
 μ increases ✓

$F_f \propto \mu$ when F_N constant ✓ (3)

QUESTION 5 MR DU TOIT

20 MARKS

5.1 (4)



1 mark for each force ✓✓✓✓

Tension force > Friction force

5.2 A force that opposes motion of an object ✓ and acts parallel to the surface the object is in contact with ✓ (2)

5.3 $F_f = \mu_k F_N$ ✓
 $= (0,5)(2)(9,8)\cos 30$ ✓
 $= 8,49 \text{ N}$ ✓ (3)

5.4 When a net force acts on an object the object accelerates in the direction of the net force. ✓ The acceleration of the object is directly proportional to the net force acting on the object and inversely proportional to the mass of the object. ✓ (2)

5.5

Block A:

$$F_{net} = ma \checkmark$$

$$40 - T = 3a \checkmark$$

Block B:

$$F_{net} = ma$$

$$T - (2)(9,8)\sin 30 - 8,49 = 2a$$

$$T = 2a + (2)(9,8)\sin 30 \checkmark + 8,49 \checkmark \text{ coe 5.3}$$

$$40 - (2a + (2)(9,8)\sin 30 + 8,49) = 3a$$

$$40 - 2a - 2(9,8)\sin 30 - 8,49 = 3a$$

$$a = 4,34 \text{ m} \cdot \text{s}^{-2} \checkmark$$

OR System:

$$F_{net} = ma \checkmark$$

$$40 - 2(9,8)\sin 30 \checkmark - 8,49 \checkmark \text{ coe 5.3} = (5)a \checkmark$$

$$a = 4,34 \text{ m} \cdot \text{s}^{-2} \checkmark \quad (5)$$

5.6

$$40 - T = 3(4,34) \checkmark \text{ coe 5.5}$$

$$T = 26,97 \text{ N} \checkmark \quad (2)$$

5.7 Increases ✓✓ (2)

QUESTION 6 MRS GOVENDER**24 MARKS**

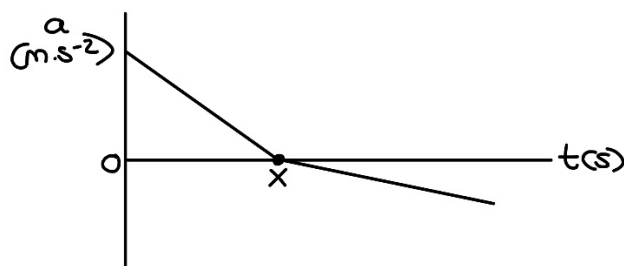
6.1.1 The product of the displacement✓ and the component of the force parallel to the displacement✓ (2)

6.1.2 Positive work: object gains energy✓
Negative work: object loses energy✓ (2)

6.1.3 Frictional force ✓ (1)

6.1.4 $a_{up} > a_{down}$ ✓
 $F_{net} \propto a$ ✓
Fnet greater on the way up as friction and Fg parallel are in the same direction✓ (3)

6.1.5



✓ Negative gradients
✓ X marked at zero velocity
✓ Gradient steeper on the way up than on the way down (3)

6.2.1 Normal force ✓
Weight ✓
Perpendicular to displacement ✓ (3)

6.2.2 $Ek = \frac{1}{2}mv^2$ ✓
 $Ek = \frac{1}{2}(70)(4)^2$ ✓
 $Ek = 560 \text{ J}$ ✓ (3)

6.2.3 The work done by a net force on an object ✓ is equal to the change in kinetic energy of the object ✓ (2)

6.2.4 $W_{net} = \Delta Ek$ ✓
 $(234,63 - 190) \times (18) = \frac{1}{2}(70)v^2 - 560$ ✓ **cof 6.2.2**
 $v = 6,24 \text{ m.s}^{-1}$ ✓ (5)

QUESTION 7 MRS GOVENDER**27 MARKS**

- 7.1.1 A material's opposition to the flow of electrical current ✓✓AON (2)
- 7.1.2 $\frac{1}{R_p} = \frac{1}{4} + \frac{1}{12} \checkmark = \frac{1}{3} \checkmark$
 $R_p = 3 \Omega \checkmark$
 $R_{ext} = 3 + 5 = 8 \Omega \checkmark$ (4)
- 7.1.3 $P = I^2 R \checkmark$
 $9 = I^2 4 \checkmark$
 $I = 1,5 A \checkmark$ (4)
- 7.1.4 $V = IR \checkmark$
 $V = (1,5)(4) \checkmark$
 $V = 6 V \checkmark$ coe 7.13 (3)
- 7.1.5
 $16 - 6 = 10 V \checkmark$ coe 7.14 OR $V = IR$
 $V = IR$ $16 \checkmark = I(8) \checkmark$ coe 7.1.2
 $10 = I(5) \checkmark$ $I = 2 A \checkmark$
 $I = 2 A \checkmark$ (3)
- 7.1.6 Remains the same ✓ (1)
- 7.1.7 Increases ✓
 $V \propto I \checkmark$ (*R is constant*) ✓
I increases through 4 ohm ✓ (4)
- 7.2.1 The rate at which work is done (or the rate at which energy is transferred) ✓✓AON (2)
- 7.2.2 L is brighter than K in circuit 1 ✓, as power is inversely proportional to resistance for constant voltage ✓

K is brighter than L in circuit 2 ✓, as power is directly proportional to resistance for constant current ✓ (4)

QUESTION 8 MR MCCOY

26 MARKS

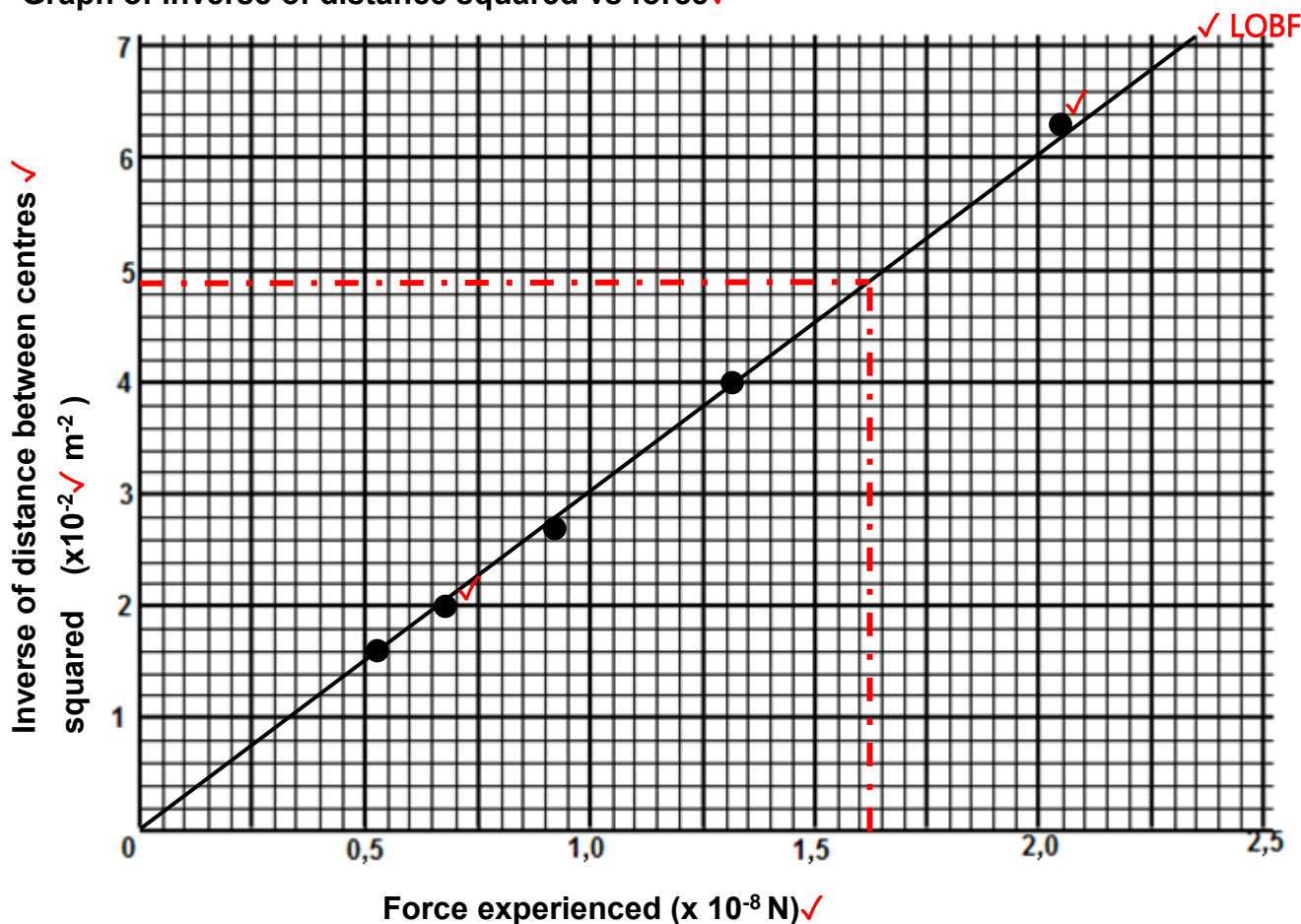
8.1.1 ANSWER SHEET. 0,518✓✓

(2)

8.1.2 ANSWER SHEET 0,016✓✓

(2)

8.2 Graph of inverse of distance squared vs force✓



(7)

8.3 The Force experienced is inverse proportional✓ to the distance between centres squared✓ order

(2)

$$8.4 \quad \frac{1}{4,5^2} = 0,049 \rightarrow 4,9 \times 10^{-2} \checkmark \checkmark = 1,65 \times 10^{-8} \text{ N} \checkmark$$

(3)

$$8.5 \quad \text{gradient} = \frac{\Delta y}{\Delta x}$$

$$= \frac{0,018 - 0 \checkmark}{0,6 \times 10^{-8} \checkmark - 0}$$

$$= 3 \times 10^6 \checkmark \text{ m}^{-2} \cdot \text{N}^{-1} \checkmark$$

(4)

$$8.6 \quad \text{gradient} = \frac{r^{-2}}{F} = \frac{1}{F \cdot r^2} \checkmark$$

$$\text{Equation: } F = \frac{Gm_1m_2}{r^2} = \frac{1}{F \cdot r^2} = \frac{1}{Gm_1m_2} \checkmark$$

$$\therefore \text{gradient} = \frac{1}{Gm_1m_2} \checkmark$$

(3)

$$8.7 \quad 3 \times 10^6 = \frac{1}{Gm_1m_2}$$

$$3 \times 10^6 \checkmark = \frac{1}{6,7 \times 10^{-11} \cdot x^2 \checkmark} \quad x = \pm 70 \text{ kg} \checkmark$$

(3)

QUESTION 9 MR MCCOY**13 MARKS**

9.1 3 ✓

(1)

9.2

$$\Delta E = hf \quad \checkmark \checkmark$$
$$(13,6 - 3,4)(1,6 \times 10^{-19}) = 6,6 \times 10^{-34} f \quad \checkmark$$
$$(10,2)(1,6 \times 10^{-19}) = 6,6 \times 10^{-34} f$$
$$f = 2,47 \times 10^{15} \text{ Hz} \quad \checkmark$$

(4)

9.3

$E = \frac{hc}{\lambda} \quad \checkmark$	OR	$c = f\lambda \quad (\text{for both formulae})$
$E = \frac{(6,6 \times 10^{-34})(3 \times 10^8)}{655 \times 10^{-9}} \quad \checkmark$		$3 \times 10^8 = f(655 \times 10^{-9}) \quad \checkmark$
$E = 3,02 \times 10^{-19} \text{ J} \quad \checkmark$		$f = 4,58 \times 10^{14} \text{ Hz}$
$E = 1,89 \text{ eV} \quad \checkmark$		$E = hf$
		$E = (6,6 \times 10^{-34})(4,58 \times 10^{14}) \quad \checkmark$
		$E = 3,02 \times 10^{-19} \text{ J} \quad \checkmark$
		$E = 1,89 \text{ eV} \quad \checkmark$

(4)

9.4

$$n = 3 \rightarrow n = 2 \quad \checkmark$$

Direction ✓

(2)

9.5

A free electron has energy of zero. ✓
Electrons will gain energy as they move up a level ✓

OR

A free electron has energy of zero ✓
so an electron releases energy as it moves to a lower level ✓

(2)